Ethan Hosier

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Education

Imperial College London — London, UK
MEng Computing - First Class Honours — *Oct. 2022 – June 2026*

Experience

Palantir | *Python, Spark, TypeScript, React.js* — April 2025 – September 2025
Forward Deployed Software Engineer, Intern — *London, UK*

- Rewrote a legacy data pipeline processing billions of customer transactions every hour, achieving 3.5× faster runtime through caching and optimized parallel execution across distributed Spark clusters
- Transformed the client's financial forecasting process by developing a spreadsheet engine from scratch supporting real-time multi-user collaboration, formulas, keyboard shortcuts and configurable templates
- Built an AI model router for Palantir's internal chat system with 4,000+ users, automatically selecting the optimal language model for each conversation
- Led meetings with the client's executives and lead engineers, and iterated on features directly with end users

Revolut | *Java, TypeScript, React.js, Kafka, Google Cloud Platform* — July 2024 – September 2024
Software Engineer, Intern — *London, UK*

- Developed the backend of an Authorised Data Source consolidating product and pricing-rule configurations into a single interface—used daily by the CEO and leadership team—eliminating manual database queries
- Reduced query latency by 99% and cut DB reads to near-zero by consuming product/pricing change events from a Kafka topic into a local in-process cache and serving price calculations directly from the cache instead of the DB.
- Integrated with internal services including the Legal Service (document assignment to tax configurations) and the People Platform (employee data in the admin app), ensuring seamless interaction across core business systems

Mia Marketing | *Go, Python, TypeScript, React.js, AWS, Microsoft Azure, Docker* — June 2024 – September 2024
Founder — *London, UK*

- Developed an MVP for a fully autonomous AI-driven digital marketing platform, attracting interest from over 40 different companies (some with revenues of >£1 million ARR)
- Built a scalable, microservices-based infrastructure on AWS and Azure of proxies, web crawlers, and scrapers
- Developed automated workflows for data-driven social media content creation, posting, and lead generation through personalized email outreach campaigns
- Awarded £2,500 in prize money from Imperial College London's Entrepreneurship Journey, alongside £1,000 in Azure credits from the Microsoft Startup Hub

Projects

Distributed Web Crawler & RAG Pipeline | *Go, TypeScript, AWS, Redis, Terraform, Docker* — March 2025

- Built a distributed worker framework in Go with Redis-based inter-node communication
- Used the framework to build a distributed web crawler and RAG pipeline, achieving the ability to scrape and index the entire Imperial College website in under 3 minutes
- Deployed on AWS ECS (Fargate) with infrastructure provisioned via Terraform, Lambdas, and CloudWatch metrics to enable auto-scaling based on queue size and throughput
- Integrated the indexed data into a custom AI agent framework used in a pilot project to replace personal tutor workflows with context-aware automated assistance

WACC Compiler | *Scala, TypeScript, Next.js* — January 2024 – March 2024

- Developed a compiler for the WACC programming language in Scala, performing lexical and semantic analysis, code generation, and optimization to output x86-64 assembly
- Built a web-based IDE using Next.js and Monaco Editor to compile and execute WACC code in-browser with syntax highlighting and real-time error feedback

Technical Skills

Java, Go, Python, TypeScript, Docker, AWS, React.js, Next.js, Redis, Terraform, Spark, Kafka